ModelIt!

Student Activity Pack

Why Do Things Fall Down?

The Science of Gravity and Falling

Kindergarten | NGSS K-PS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

Multiple Choice: Circle the best answer for each question.

1 What happens when you let go of a ball?

- **A.** It falls down
- **B.** It floats up
- **C.** It stays in the air

2 If you drop a rock, which way does it go?

- **A.** It goes sideways
- **B.** It goes up
- **C.** It goes down

3 What do you think makes things fall to the ground?

- **A.** The wind blows them down
- **B.** Something invisible pulls them down
- **C.** They want to be on the ground

4 If you hold a toy up high and let go, what will happen?

- **A.** It will fly away
- **B.** It will fall down
- **C.** Nothing will happen

1 What happens when you let go of a ball? (Circle: it falls down / it floats up)

2 Draw something falling down from the sky.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Drop Height | Fall Speed

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Low Drop

Drop Height set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High Drop

Drop Height set to high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do things always fall down when you let go of them? Use evidence from your simulation to support your answers.

8 STEM Challenge: Build a Ball Drop Tower

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YOUR ENGINEERING MISSION

Build a tower and test dropping balls from different levels to see how height changes how fast they fall.

THE SCENARIO

Your class has been asked to help build a fun new ball drop game for the school fair! You need to figure out which drop height makes the ball bounce the highest when it lands. Use what you learned about gravity to design the best drop.

Engineering Design Thinking

1. HOW MANY LEVELS WILL YOUR TOWER HAVE?

2. HOW WILL YOU MAKE SURE THE BALL DROPS STRAIGHT DOWN?

3. HOW WILL YOU TELL WHICH DROP MADE THE BIGGEST BOUNCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Astronauts study gravity in space where things float because there is less gravity pulling on them. They earn $66,000-$144,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 What is gravity?

- **A.** A push that makes things go up
- **B.** A pull that makes things fall down
- **C.** A sound you hear

2 When you drop a ball from HIGH up, what happens?

- **A.** It falls slowly
- **B.** It falls faster
- **C.** It does not fall at all

3 In our model, which part do WE get to decide?

- **A.** Fall Speed
- **B.** Drop Height
- **C.** Gravity

4 Does gravity ever stop pulling on things?

- **A.** Yes, at night time
- **B.** Yes, when things are sitting still
- **C.** No, it never stops